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EDUCATION

PhD of Computer Science - University of Bonn October 2024 - Present

Specialization: 3D/4D Reconstruction with Neural Fields and Gaussian Splatting

Supervisor: Dr. Zorah Löhner

Status: Current contract through October 2027 (extension anticipated)

Master of Computer Science - Technical University of Munich October 2021 - April 2024

Thesis Title: Neural Shading-based Optimization of Neural Implicit Representations

Specialization: Computer Vision and Machine Learning

Supervisor: Dr. Matthias Nießner

Thesis Grade: 4.0/4.0

Bachelor of Computer Engineering - Cairo University September 2016 - April 2021

Thesis Title: Generating Human Faces from Input Text using GANs

WORK EXPERIENCE

Doctoral Researcher — University of Bonn October 2024 - Present

Working in the Geometry in Machine Learning group on 4D Reconstruction of Deformable Objects from Multi-view RGB videos using Neural SDFs and Gaussian Splatting.

Machine Learning Working Student — Siemens (Munich, Germany) March 2023 - June 2024

Working on 3D Object Detection and Segmentation for Autonomous Driving and Robotics from RGB Images and LiDAR.

Teaching Assistant — Technical University of Munich Feb 2023 - March 2024

Teaching Assistant for Introduction to Deep Learning course and Machine Learning course.

Deep Learning Student Assistant — Fortiss (Munich, Germany) Dec 2021 - Dec 2022

Building robust object detection models for autonomous driving. Also building metrics that measure their robustness on test data. Also, worked on the visualization of the learned features for Object Detection.

Deep Learning Intern - Affectiva (Cairo, Egypt) Jun 2020 - Oct 2020

Built module for real-time Deep Learning model interpretability for facial emotion classification. The module is able to visualize the important parts of the input image that affects the model decisions.

Deep Learning Research Intern - Valeo (Cairo, Egypt) Jul 2019 - Oct 2019

Published 2 Research papers to Autonomous Driving workshops at NeurIPS19 and ICCV19 in Semantic Motion Segmentation for Autonomous Driving with Deep Learning.

PUBLICATIONS

1. M. Ebbed, Z. Löhner “**MoAngelo: Motion-Aware Neural Surface Reconstruction for Dynamic Scenes**” *International Conference on 3D Vision (3DV)*, 2026.
2. M. Ebbed, H. Rashed, A. El Sallab, S. Yogamani, “**RST-MODNet: Real-Time Spatio-Temporal Moving Object Detection for Autonomous Driving,**” *Neurips ML4AD Workshop*, 2019.
3. H. Rashed, M. Ebbed, V. Vaquero, A. El Sallab, G. Sistu, S. Yogamani, “**FuseMODNet: Real-Time Camera and LiDAR Based Moving Object Detection for Robust Low-Light Autonomous Driving,**” *IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, 2019.

SKILLS

Programming Languages:	Python, C++
Deep Learning Libraries:	Pytorch, MinkowskiEngine, Tensorflow, Keras, ONNX, TensorflowJS, TensorRT, TinyCUDA NN
Machine Learning Libraries:	scikit-learn, numpy, pandas, matplotlib
Other Technologies:	OpenCV, CMake, COLMAP, CUDA, Docker, GIT, Linux, Blender, Meshlab
Areas of experience:	3D Computer Vision, Visual Computing, 3D/4D Reconstruction, Neural Fields, Gaussian Splatting, Neural Rendering, Neural Implicit Representations, Multi-view Geometry, Generative Models

PROJECTS

3D Reconstruction for Dynamic Scenes from Multi-view Videos

Developed a novel framework extending NeuralAngelo for dynamic 3D scene reconstruction from multi-view videos. The method introduces a flexible template-based approach with jointly optimized deformation fields, enabling high-quality geometry tracking and refinement across time. Achieved state-of-the-art reconstruction accuracy on the ActorsHQ dataset compared to existing dynamic reconstruction methods. **(Python, PyTorch) (3DV 2026)**

Neural shading-based optimization of signed distance functions — Master Thesis

Developed a novel approach for 3D reconstruction leveraging neural fields, facilitating the joint optimization of a neural Signed Distance Function (SDF), material, and lighting using RGB images with known camera poses. Material properties are characterized by diffuse term and optionally include a specular term, while lighting is represented by either an environment map or spatially varying spherical harmonics coefficients. Our method demonstrates enhanced reconstruction quality due to the appearance decomposition into material and lighting, enabling the incorporation of finer details into the reconstructed surfaces. **(Python, PyTorch)**

ShadedSDF: Volumetric surface reconstruction with appearance decomposition constraints

Developed a technique to integrate appearance decomposition constraints (shading cues), leveraging inverse rendering, to refine neural Signed Distance Functions (SDFs). Our method enhances state-of-the-art 3D reconstruction methods such as MonoSDF by incorporating the shading cues, thereby improving the quality of the resulting reconstructions. **(Python, PyTorch) (Submitted to SIGGRAPH ASIA 2023)**

SparseVoxels: Learning Sparse Implicit Features For Novel View Synthesis

Developed a Deep Learning approach that surpasses Nerf in Novel-view synthesis while achieving a remarkable speedup of **35 times**. The method begins by optimizing a Sparse Implicit Voxel Grid for the scene using RGB images with known camera poses. Subsequently, the Voxel Grid is projected onto a 2D plane and inputted into a Rendering Convolutional Neural Network (CNN) for the synthesis of novel views. (**Python, PyTorch, MinkowskiEngine, COLMAP**)

Inverse Rendering using Neural Implicit Representations

Built a Deep Learning method that is capable of reconstructing a relightable mesh from provided RGB images of an object with known camera poses. Our approach utilizes neural fields to represent geometry, appearance, material properties, and environmental lighting. Remarkably, the method achieves state-of-the-art results, surpassing renowned baselines for inverse rendering such as Nerfactor, NerV, and NerD.. (**Python, PyTorch, COLMAP**)

Realtime Dense Surface Mapping and Tracking

Built a System with CUDA, C++, and OpenCV that performs real-time surface reconstruction and camera tracking for a scene from given input depth Images. (**C++, OpenCV, CUDA**)

Neural 3D shapes Generation and Completion

Built a Deep Learning method that performs shape completion and generation from an input signed distance function (SDF) grid for the shape. The SDF grid is processed through 3D CNN and then a Graph Convolutional Neural Network is used to predict the triangular mesh for the object. (**Python, PyTorch**)

Semantic Motion Segmentation For Autonomous Driving

Built a Deep Learning System that does Sensor Fusion of RGB images and Optical flow to improve Semantic Motion Segmentation for Autonomous Driving. It utilizes also temporal information to furtherly improve the results. The project achieves state-of-art results and was published in Autonomous driving workshops in NeurIPS19 and ICCV19. (**Python, PyTorch, Tensorflow**)

Ritratista — B.Sc graduation project

Built a Deep Learning-based System that extends StyleGAN2 to be conditioned on input textual description of the face. It also enables the user to change the attributes and head pose of the generated face. (**Python, PyTorch**)